POLITEKNIK MALAYSIA
TUN SYED NASIR

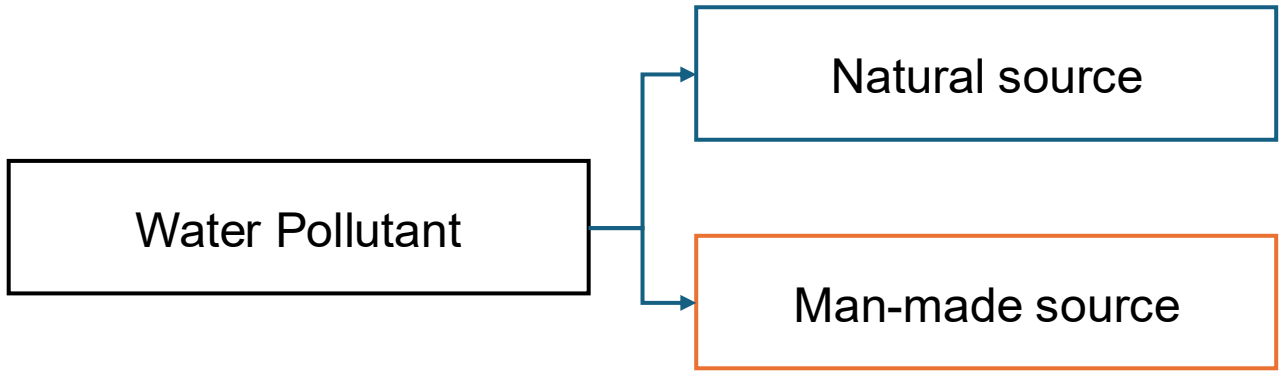


CHAPTER 3: WASTEWATER CONTROL | Ts. MUHAMMAD AZRI IZANI BIN MOHAMAD HALIM

INTRODUCTION

Wastewater is defined as used water from any combination of domestic, industrial, commercial or agricultural activities, surface runoff or stormwater, and any sewer inflow or sewer infiltration.

Sources of water pollutant:



Example of water pollutant:

1. Ammonia
2. Organic Matter
3. Nutrient
4. Microorganism
5. Colour
6. Odor
7. Heavy Metal
8. Suspended solid

WASTEWATER PARAMETER

- Wastewater Quality Analysis Parameter:

CATEGORY	PARAMETER	FULL NAME
Organics (Determination of the concentration of carbon-based compounds)	BOD	Biochemical Oxygen Demand
	COD	Chemical Oxygen Demand
	TOC	Total Organic Carbon
	F,O&G	Fat, Oil and Grease
Solids (Measurement of the concentration of particulate solids that can dissolve or suspend in wastewater)	TS	Total Solids
	TSS	Total Suspended Solids
	TDS	Total Dissolved Solids
	TVS	Total Volatile Solids
	TFS	Total Fixed Solids
Nutrients (A measurement of the concentration of targeted nutrients that can contribute to the acceleration of eutrophication)	NH_3	Ammonia
	TKN	Total Kjeldahl Nitrogen
	N-N	Nitrite/Nitrate Nitrogen
	TP	Total Phosphorus
Physical Properties (A measure a varied group of constituents directly impact wastewater treatability)	Temperature	Temperature
	Color	Color
	Turbidity	Turbidity
	Odor	Odor

WASTEWATER PARAMETER

- The quality of surface water in Malaysia is assessed using the **Water Quality Index (WQI)**.
- The WQI combines several water quality parameters, such as dissolved oxygen, biochemical oxygen demand (BOD), chemical oxygen demand (COD), ammoniacal nitrogen, suspended solids, and pH, to determine the overall condition of rivers and other surface waters.
- Based on the WQI value, water bodies are classified into different categories ranging from clean to polluted.
- The regulatory authority responsible for controlling wastewater effluent quality in Malaysia is the Department of Environment (DOE). The DOE enforces environmental protection through the Environmental Quality Act 1974 and its related regulations.

ENVIRONMENTAL QUALITY ACT 1974

- Effluent discharge standard for Industrial Effluent and Crude Palm Oil:

ACCEPTABLE CONDITIONS FOR DISCHARGE OF INDUSTRIAL EFFLUENT OR MIXED EFFLUENT OF STANDARDS A AND B

	Parameter	Unit	Standard A	Standard B
	(1)	(2)	(3)	(4)
(i)	Temperature	°C	40	40
(ii)	pH Value	–	6.0-9.0	5.5-9.0
(iii)	BOD_5 at 20°C	mg/L	20	50
(iv)	Suspended Solids	mg/L	50	100
(v)	Mercury	mg/L	0.005	0.05
(vi)	Cadmium	mg/L	0.01	0.02
(vii)	Chromium, Hexavalent	mg/L	0.05	0.05
(viii)	Chromium, Trivalent	mg/L	0.20	1.0
(ix)	Arsenic	mg/L	0.05	0.10
(x)	Cyanide	mg/L	0.05	0.10
(xi)	Lead	mg/L	0.10	0.5
(xii)	Copper	mg/L	0.20	1.0
(xiii)	Manganese	mg/L	0.20	1.0
(xiv)	Nickel	mg/L	0.20	1.0
(xv)	Tin	mg/L	0.20	1.0
(xvi)	Zinc	mg/L	2.0	2.0
(xvii)	Boron	mg/L	1.0	4.0
(xviii)	Iron (Fe)	mg/L	1.0	5.0
(xix)	Silver	mg/L	0.1	1.0
(xx)	Aluminium	mg/L	10	15
(xxi)	Selenium	mg/L	0.02	0.5
(xxii)	Barium	mg/L	1.0	2.0
(xxiii)	Fluoride	mg/L	2.0	5.0
(xxiv)	Formaldehyde	mg/L	1.0	2.0
(xxv)	Phenol	mg/L	0.001	1.0
(xxvi)	Free Chlorine	mg/L	1.0	2.0
(xxvii)	Sulphide	mg/L	0.50	0.50
(xxviii)	Oil and Grease	mg/L	1.0	10
(xxix)	Ammoniacal Nitrogen	mg/L	10	20
(xxx)	Colour	ADMI*	100	200

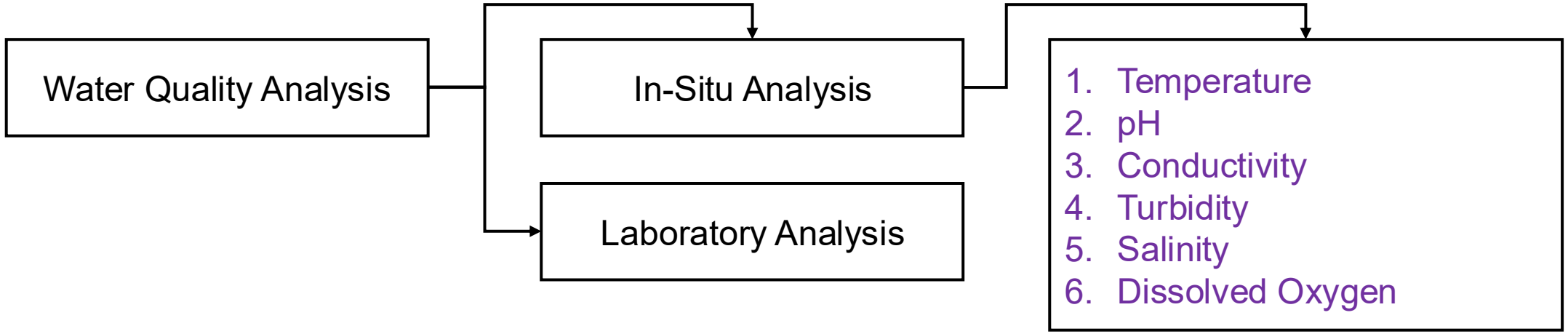
*ADMI–American Dye Manufacturers Institute

Parameters	DOE Discharge Limit (1986 onwards)[a]	Environmental Quality Act[b]
BOD_3 (mg/l)	50 (20)[d]	100
COD (mg/l)	1000	1000[c]
Total Solids (mg/l)	1500	1500[c]
Suspended Solids (mg/l)	400	400
Oil and grease (mg/l)	50	50
Ammoniacal Nitrogen (mg/l)	100	150[c]
Total Nitrogen (mg/l)	200	200[c]
pH	5.0	5.0 – 9.0
Temperature (°C)	45	45

a Malaysian Department of Environment (DOE)
b Parameters Limit of Environmental Quality (Prescribed Premises) (Crude Palm Oil) (Amendment) Regulation 1997
c No new value stipulated since 1982
d Stringent standards in environmentally sensitive areas of Sabah and Sarawak, Malaysia starting 2006

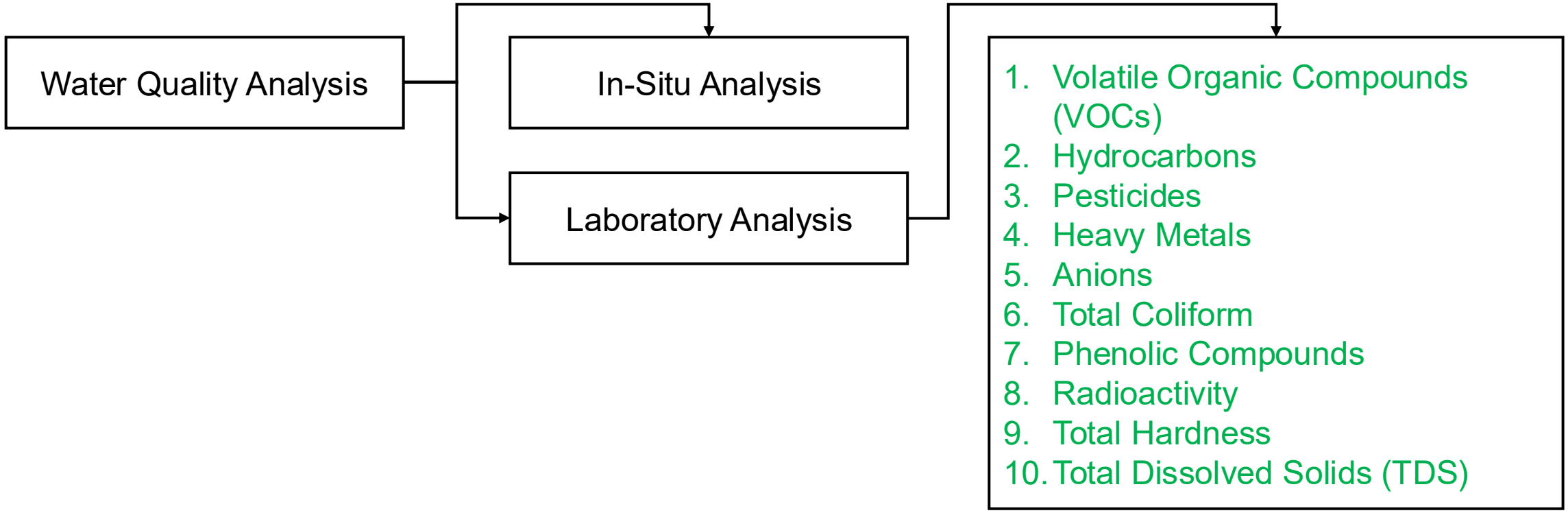
WATER QUALITY ANALYSIS APPROACH

- Water quality analysis can be divided into two main approaches: In-Situ Analysis and Laboratory Analysis.



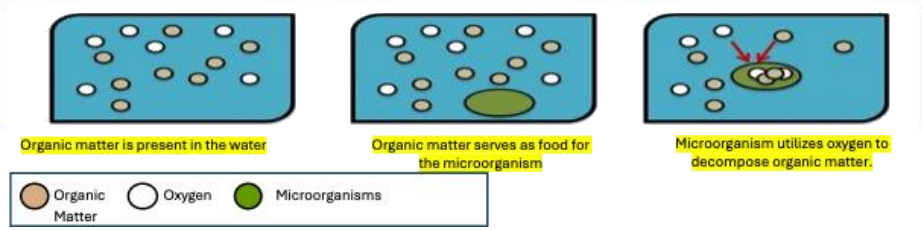
WATER QUALITY ANALYSIS APPROACH

- Water quality analysis can be divided into two main approaches: In-Situ Analysis and Laboratory Analysis.



BIOCHEMICAL OXYGEN DEMAND (BOD)

- Definition: Biochemical Oxygen Demand (BOD) is the amount of dissolved oxygen required by microorganisms (mainly bacteria) to decompose and oxidize biodegradable organic matter present in water under aerobic conditions.

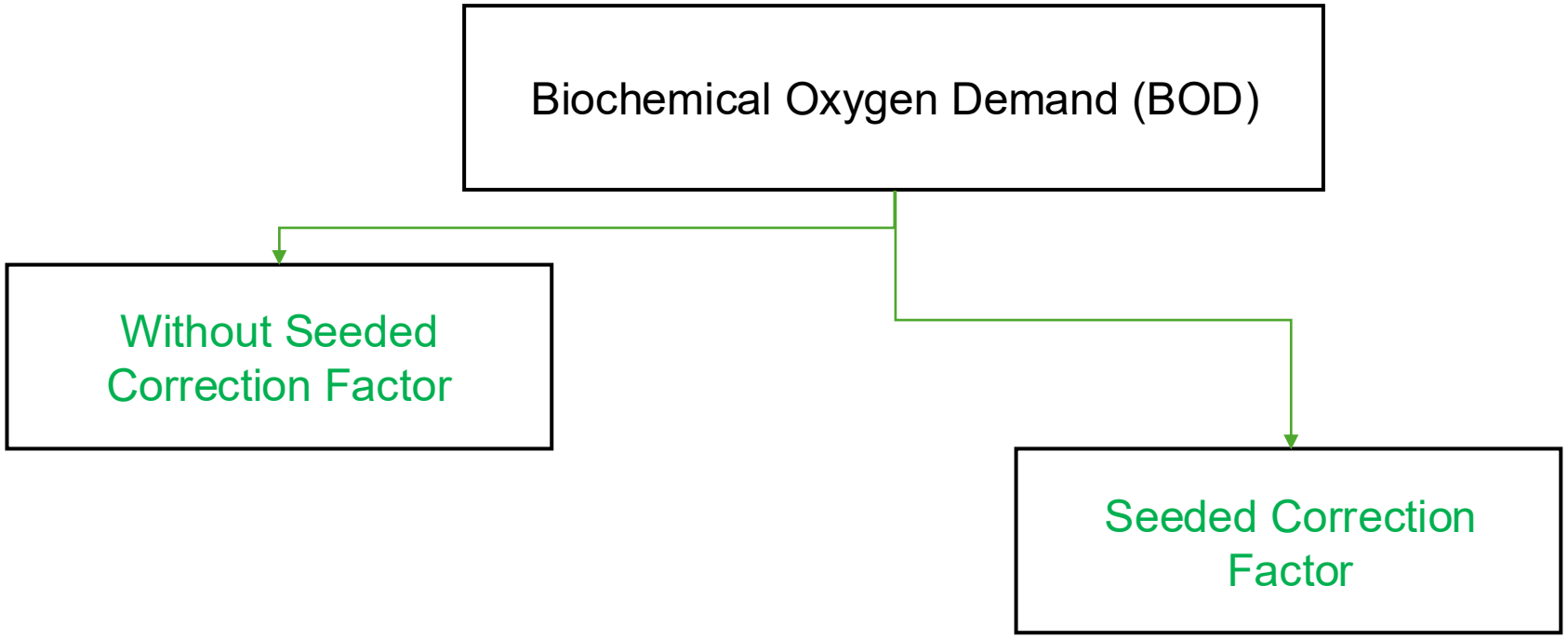


Key Characteristics of BOD:

1. Measured in milligrams per litre (mg/L) of oxygen.
2. Indicates the amount of biodegradable organic pollution in water.
3. Depends on the activity and efficiency of microorganisms.
4. The oxidation process is relatively slow because it relies on biological reactions.
5. Widely used in wastewater treatment plants to evaluate wastewater strength and treatment efficiency.

BIOCHEMICAL OXYGEN DEMAND (BOD)

- Biochemical Oxygen Demand (BOD) calculation:



BIOCHEMICAL OXYGEN DEMAND (BOD)

Aspect	Without Seed Correction	With Seed Correction
Purpose	To measure BOD assuming all oxygen consumption is due to organic matter in the sample.	To measure true BOD by removing oxygen demand contributed by added microorganisms (seed).
When Applied	Used for samples with sufficient natural microorganisms (e.g., domestic sewage, raw wastewater).	Used for samples with low microbial activity or toxic/industrial wastewater requiring added seed.
Accuracy	Less accurate when seed is present because seed oxygen demand is included in result.	More accurate because seed oxygen demand is corrected and removed.
Complexity	Simple calculation using only DO values and dilution factor.	More complex because it includes seed control correction factor (f).
BOD Value Obtained	Usually higher (may be overestimated).	Usually lower and more realistic for actual organic pollution.

BOD WITHOUT SEEDED CORRECTION

- Biochemical Oxygen Demand (BOD) calculation without seeded correction factor:

$$BOD_5 = \frac{DO_i - DO_f}{P}$$

$$P = \frac{\text{Volume of wastewater (Raw), mL}}{\text{Total Volume (Wastewater + Dilution water), mL}}$$

DO_i : Initial Dissolve Oxygen, mg/L

DO_f : Final Dissolve Oxygen, mg/L

P : Dilution factor

EXAMPLE 1

1. Calculate dilution factor, P if 100 mL of raw wastewater sample is diluted to 300 mL total sample volume.
2. Calculate dilution factor, P for BOD testing if raw wastewater sample is 100 mL mix with dilution water of 400 mL.
3. Calculate BOD_5 value for both cases if initial DO reading is 6 mg/L and final DO reading is 4 mg/L.

BOD WITH SEEDED CORRECTION

- Biochemical Oxygen Demand (BOD) calculation with seeded correction factor:

$$BOD_5 = \frac{(DO_i - DO_f) - (B_i - B_f)f}{P}$$

$$f = 1 - P$$

DO_i	Initial dissolved oxygen of diluted wastewater (mg/L)
DO_f	Final dissolved oxygen after 5 days (mg/L)
B_i	Initial DO of seed control bottle (mg/L)
B_o	Final DO of seed control bottle (mg/L)
P	Dilution factor of wastewater sample
f	Seed correction factor

EXAMPLE 2

- Calculate Biochemical Oxygen Demand (BOD_5) value if 15 mL of raw wastewater sample is diluted to 300 mL of total sample volume. Given:

DO_i	8.8 mg/L
DO_f	1.9 mg/L
B_i	9.1 mg/L
B_o	7.9 mg/L

CHEMICAL OXYGEN DEMAND

- **Definition:** Chemical Oxygen Demand (COD) is the amount of oxygen needed to chemically oxidize the organic (wastes) in water sample by using a strong chemical oxidizing agent.
- Some organic compounds are difficult or impossible for microorganisms to degrade, such as:
    1. Cellulose
    2. Phenols
    3. Benzene
    4. Tannic acid
    5. Certain pesticides
    6. Various industrial chemicals
- Since BOD relies on microbial activity, these compounds may not be fully measured by BOD. COD uses a chemical oxidant instead, allowing a more complete measurement of the oxygen-demanding substances present.

CHEMICAL OXYGEN DEMAND*

- These are the COD key characteristics:

Aspect	Description
Principle	Organic matter is oxidized chemically using a strong oxidizing agent (commonly potassium dichromate, $K_2Cr_2O_7$).
Test Method	Water sample is digested with a strong oxidizing agent in acidic conditions at high temperature for a specified period. The amount of oxidant consumed is measured.
Disadvantages	1. Does not indicate the rate of biological degradation. 2. Cannot distinguish between biodegradable and biologically inert organic matter. 3. Provides less information about the actual environmental impact compared to BOD.
Test Duration	Approximately 2 to 4 hours.
Dependence on Microorganisms	No.

COD VS BOD

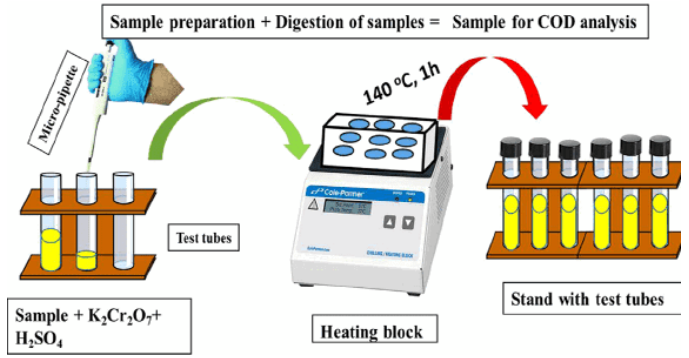
- BOD and COD characteristics comparison:

Characteristic	BOD (Biochemical Oxygen Demand)	COD (Chemical Oxygen Demand)
Definition	Amount of oxygen required by microorganisms to biologically decompose biodegradable organic matter.	Amount of oxygen required to chemically oxidize organic matter in water or wastewater.
Oxidation Process	Biological oxidation	Chemical oxidation
Oxidizing Agent	Microorganisms (aerobic bacteria)	Strong chemical oxidizing agents (e.g., potassium dichromate)
Test Method	Water sample is incubated in a sealed bottle for 5 days at 20°C. The decrease in dissolved oxygen (DO) is measured to determine oxygen consumption by microorganisms.	Water sample is digested with a strong oxidizing agent in acidic conditions at high temperature for a specified period. The amount of oxidant consumed is measured.

COD VS BOD

- BOD and COD characteristics comparison:

Characteristic	BOD (Biochemical Oxygen Demand)	COD (Chemical Oxygen Demand)
Test Duration	Slow (5 days)	Fast (approximately 2 to 4 hours)
Sensitivity to Toxic Substances	High: Toxic compounds may inhibit microbial activity and affect results.	Low: Not depend on microbial activity.
Typical Value	Lower than COD	Higher than BOD
Major Limitation	Long testing period and affected by toxic substances.	Cannot distinguish between biodegradable and non-biodegradable organic matter.



EXAMPLE 3

- Explain COD and BOD differences according to oxidizing agent, test method, test duration and major limitation.

SOLIDS

- Definition: Measure of all the suspended, colloidal, and dissolved solids in a sample of water.

Sources of total solids

Soil erosion: It will introduce suspended particles such as clay and silt (TSS), and dissolved minerals such as halite (NaCl) and limestone ($CaCO_3$) (TDS).

Agricultural runoff: Contains fertilizers, pesticides, and suspended soil particles that increase both TDS and TSS.

Industrial waste: Contribute dissolved salts, particulates, and wastewater effluents, increasing total solids in water bodies.



IMPACT OF HIGH TOTAL SOLID

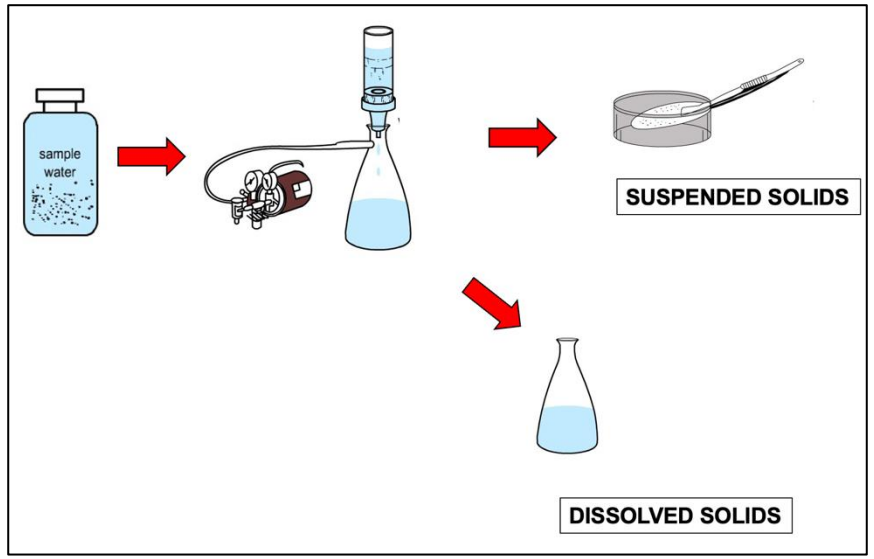
- These are the impact of high total solid content to environment:

Parameter	Impact on Water Quality and Aquatic Life
High Total Suspended Solids (TSS)	Reduces water clarity (turbidity), limiting sunlight penetration into the water. This decreases the rate of photosynthesis by aquatic plants and algae, leading to lower dissolved oxygen levels.
High Total Dissolved Solids (TDS)	Increases conductivity, salinity, alkalinity, and water hardness. Elevated salinity can create osmotic stress, making it difficult for aquatic organisms to survive and adapt.
Overall Effect	Degrades water quality, disrupts aquatic ecosystems, and may affect the suitability of water for drinking, irrigation, and industrial use.

TOTAL SOLID

- Total solid calculation:

> Total solid = Total dissolve solid (TDS), mg/L + Total suspended solid (TSS), mg/L



EXAMPLE 4

- Calculate Total Solid, Total Suspended Solid (TSS) and Total Dissolved Solid (TDS) if the volume of the sample for each analysis was fixed at 800 ml.

NO	DESCRIPTION	WEIGHT (g)
1	Weight of the clean evaporating dish	52.357
2	Weight of the dish and the residue after 105°C	53.098
3	Weight of the clean filter paper	19.720
4	Weight of the filter paper and the residue 105°C	20.371

PHOSPHORUS

- Phosphorus in natural water and wastewater is mainly present as phosphate forms, because phosphorus is highly reactive and does not exist freely for long.
- Phosphorus enters water systems from both natural and human activities:
    1. Natural processes: decomposition of organic matter, soil erosion, biological cycles.
    2. Domestic wastewater: detergents, cleaning agents, human waste.
    3. Industrial discharge: food processing, oil and fat industry, fertilizer-related processes.
- High phosphorus in wastewater can cause:
    1. Eutrophication (excess algal growth) May block out sunlight preventing other photosynthetic organisms to survive and cause oxygen depletion in water bodies.
    2. Cause a risk to human health from threat near to drinking water intakes.
    3. Decrease the aesthetic appearance of water bodies.

PHOSPHORUS

- Phosphorus (Phosphate) Analysis using the **Ascorbic Acid Method**:
    1. Digest the sample to convert all forms of phosphorus into phosphate (PO_4^{3-}).
    2. Add acid to create the correct acidic condition for the reaction.
    3. Add molybdate and antimony reagents. The phosphate reacts with these reagents to form a phosphomolybdate complex.
    4. Add ascorbic acid (reducing agent). This converts the complex into a blue-colored compound.
    5. Allow the blue color to develop completely.
    6. Measure the absorbance of the blue color using a spectrophotometer, usually at 880 nm.
    7. Calculate the phosphorus concentration. The darker the blue color, the higher the phosphorus concentration.

SULPHATE

- Definition: Sulfate (SO_4^{2-}) is a naturally occurring substance found in water. Its concentration can increase due to both natural processes and human activities.
- Sulfate (SO_4^{2-}) is commonly found in wastewater from the oil and fats industry, particularly during the oil refining process. One major source is the soapstock acidulation process, where sulfuric acid (H_2SO_4) is added to soapstock to recover fatty acids. As a result, sulfate ions are generated and discharged into wastewater.
- Impact of excessive sulphate toward environment:
    1. Increase TDS level and makes water less suitable for drinking and industrial use.
    2. High sulfate can disturb balance of aquatic ecosystems and reduce survival of sensitive organisms.
    3. Sulfate reduction processes depleting oxygen in water bodies, making water unsuitable for fish and aquatic organisms.
    4. Animals like cattle convert sulfate to hydrogen sulfide in digestion, which can be toxic to them

SULPHATE

- Sulphate Determination **($BaCl_2$ Turbidimetric Method):**
    1. Sulfate analysis is performed by preparing standard solutions (10–50 mg/L), a blank, and an unknown water sample.
    2. Each flask is added with 5 mL of conditioning reagent to stabilize the reaction and reduce interference.
    3. Distilled water is added until the total volume reaches 100 mL for uniformity.
    4. Barium chloride ($BaCl_2$) is added, reacting with sulfate ions to form white $BaSO_4$ precipitate, producing turbidity.
    5. Turbidity is measured using a UV-Vis spectrophotometer, where higher turbidity indicates higher sulfate concentration.
    6. A calibration curve is prepared using standard solutions by plotting turbidity or absorbance against concentration.
    7. The sulfate concentration of the unknown sample is obtained by comparing its reading with the calibration curve.

EXAMPLE 5

1. State 2 impacts of excessive sulphate towards environment.

2. State 3 activities that introduce phosphorus to the water systems from both natural and human activities.

WASTEWATER GENERATION IN O&F INDUSTRY

- Process/Stage that affecting wastewater generation in oil and fats industry:

1. Soapstock Handling
2. Deodorization process
3. Refining (Wash / Separation Stage)
4. Clean-up and Handling Losses process

WASTEWATER GENERATION IN O&F INDUSTRY*

Factor	Source of Wastewater	Key Pollutants	Wastewater Characteristics
Soapstock Handling	Separation and treatment of soapstock from neutralisation process	1. Free fatty acids (FFA) 2. Soaps 3. Phosphates 4. Sulphates 5. Residual NaOH Emulsified oil	1. High COD and BOD 2. High alkalinity 3. Oily effluent 4. Strong organic load
Deodorization	Condensate from steam stripping and vacuum systems	1. Volatile fatty acids 2. FOG (fat, oil, grease) 3. Odor compounds 4. Condensate Contaminants	1. Low volume but very high strength wastewater. 2. High oil content 3. Warm effluent

WASTEWATER GENERATION IN O&F INDUSTRY

Factor	Environmental Impact	Control / Management Measures
Soapstock Handling	1. Major contributor to organic pollution 2. Increases oxygen demand in receiving water	1. Acidulation to recover fatty acids 2. Selling crude soapstock 3. Reuse as feed additive (Fat Additive) 4. Proper separation and dewatering
Deodorization	1. High localized pollution potential 2. Contributes to FOG contamination and odor issues	1. Use of indirect heat exchangers 2. Recycle condensate 3. Improved separator systems (barometric/skimmer) 4. Reduce steam losses

WASTEWATER GENERATION IN O&F INDUSTRY*

Factor	Source of Wastewater	Key Pollutants	Wastewater Characteristics
Refining (Washing & Separation Stage)	1. Wash water discharge 2. Centrifuge effluent	1. Emulsified oil 2. Suspended solids 3. Neutral oil 4. Trace chemicals	1. Moderate to high TSS Emulsified oil 2. Variable flow
Clean-up & Handling Losses	General plant wastewater streams from housekeeping and losses	1. FOG 2. Dust particles 3. Suspended solids 4. Residual oils	1. Highly variable wastewater quality 2. Intermittent but frequent low-to-medium strength pollution

WASTEWATER GENERATION IN O&F INDUSTRY

Factor	Environmental Impact	Control / Management Measures
Refining (Washing & Separation Stage)	1. Increases turbidity 2. Oil contamination in water bodies 3. Reduces oxygen transfer in aquatic systems	1. Improve centrifuge efficiency 2. Process monitoring 3. Prevent backpressure issues 4. Use secondary separation (gravity/DAF)
Clean-up & Handling Losses	1. Cumulative pollution load 2. Often underestimated source of contamination	1. Good housekeeping 2. Leak prevention 3. Regular maintenance 4. Wastewater auditing 5. Loss minimization program

EXAMPLE 6

1. State 4 process/stage that affecting wastewater generation in oil and fats industry.

2. State 2 environmental impact from refining process wastewater.

THANK YOU

PLEASE SUBMIT YOUR MIND MAP